WIZ − ECA, all features (H_Shannon, H_Renyi, H_Tsallis: lower; others: upper)

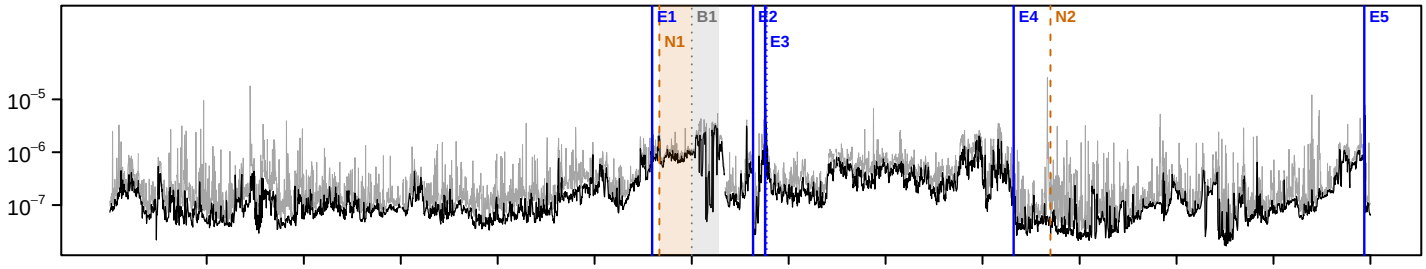
Blue: eruptions E1−E5 (line = onset, band = eruptive period) Grey bands: precursor window of the best setting (dT weeks, ending tau weeks before each onset)

Orange dashed: left out of the test, grey dotted: not treated as eruptions (N1 2012−09−01 to 12−31, B1 2013−01−01 to 04−10, N2 2016−09−13)

Red: windows beyond the best threshold (dark = inside a precursor window) Dashed: the four percentile thresholds (bold = best)

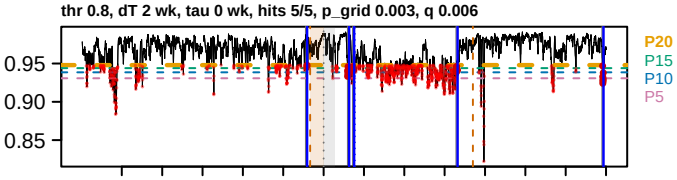
WIZ RSAM (10−minute values, log scale): black = daily median, grey = daily maximum

Eruptions counted (UTC, scenario 'definite'): E1 2012−08−04, E2 2013−08−19, E3 2013−10−04 to 10−11, E4 2016−04−27, E5 2019−12−09



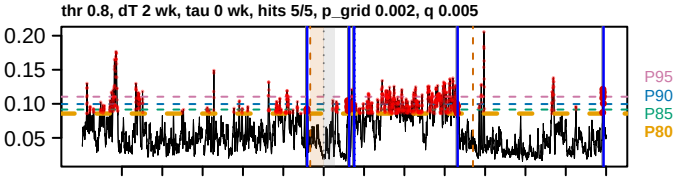
H_Shannon [lower]

thr 0.8, dT 2 wk, tau 0 wk, hits 5/5, p_grid 0.003, q 0.006



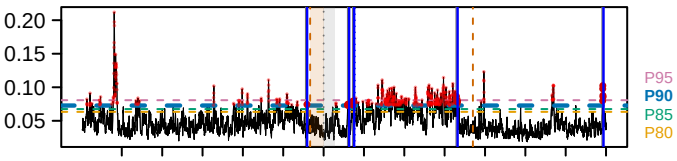
C_Shannon [upper]

thr 0.8, dT 2 wk, tau 0 wk, hits 5/5, p_grid 0.002, q 0.005



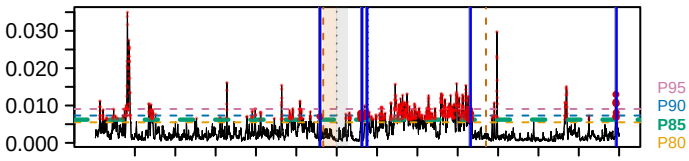
Fisher_Info [upper]

thr 0.9, dT 2 wk, tau 0 wk, hits 5/5, p_grid 0.002, q 0.005



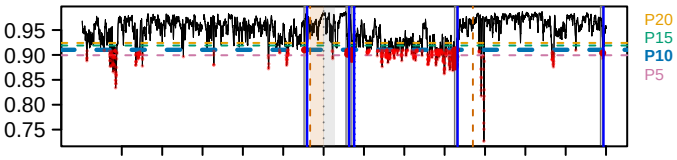
C_Fisher [upper]

thr 0.85, dT 2 wk, tau 0 wk, hits 5/5, p_grid 0.003, q 0.006



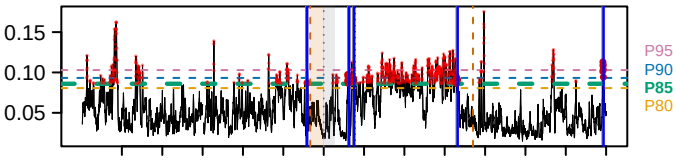
H_Renyi [lower]

thr 0.9, dT 3 wk, tau 1 wk, hits 5/5, p_grid 0.005, q 0.008



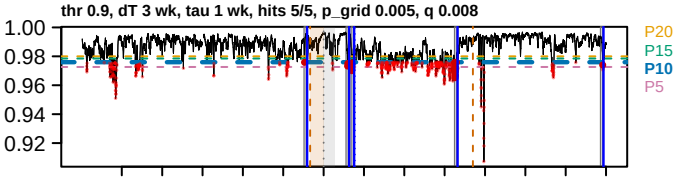
C_Renyi [upper]

thr 0.85, dT 2 wk, tau 0 wk, hits 5/5, p_grid 0.002, q 0.005



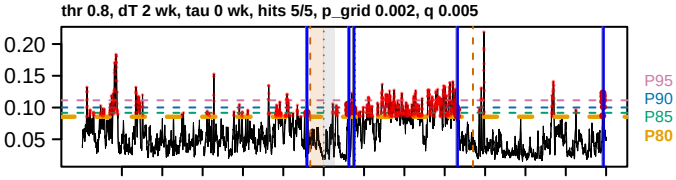
H_Tsallis [lower]

thr 0.9, dT 3 wk, tau 1 wk, hits 5/5, p_grid 0.005, q 0.008



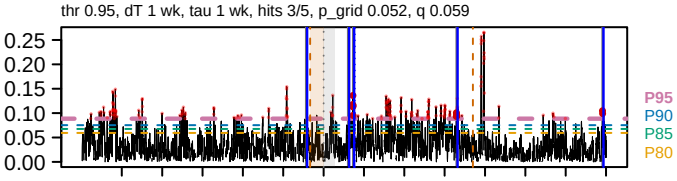
C_Tsallis [upper]

thr 0.8, dT 2 wk, tau 0 wk, hits 5/5, p_grid 0.002, q 0.005



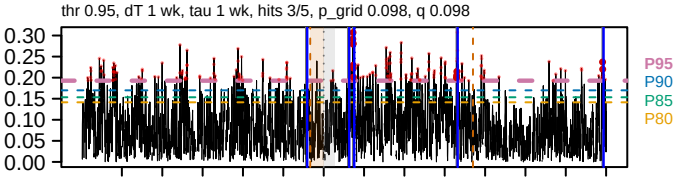
Var_H_Shannon [upper]

thr 0.95, dT 1 wk, tau 1 wk, hits 3/5, p_grid 0.052, q 0.059



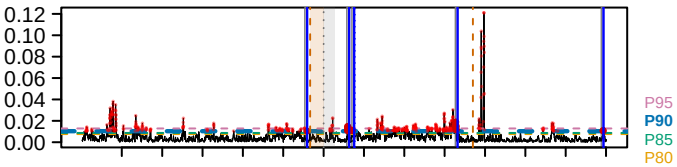
Var_C_Shannon [upper]

thr 0.95, dT 1 wk, tau 1 wk, hits 3/5, p_grid 0.098, q 0.098



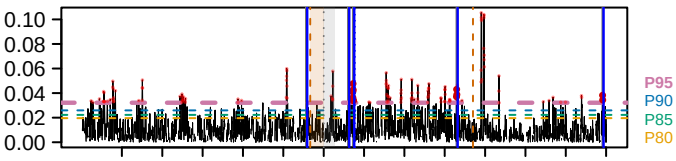
Var_H_Renyi [upper]

thr 0.9, dT 3 wk, tau 0 wk, hits 5/5, p_grid 0.007, q 0.009



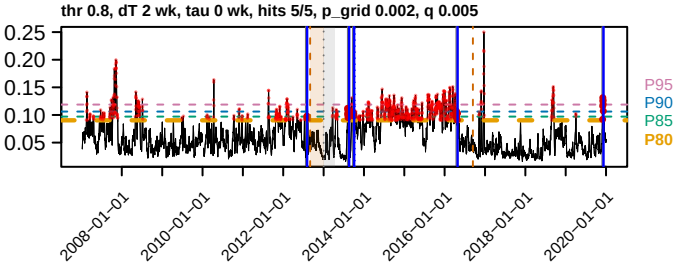
Var_H_Tsallis [upper]

thr 0.95, dT 1 wk, tau 1 wk, hits 3/5, p_grid 0.054, q 0.059



Disequilibrium [upper]

thr 0.8, dT 2 wk, tau 0 wk, hits 5/5, p_grid 0.002, q 0.005



date (UTC); every feature value is plotted at the end of its 7−day window